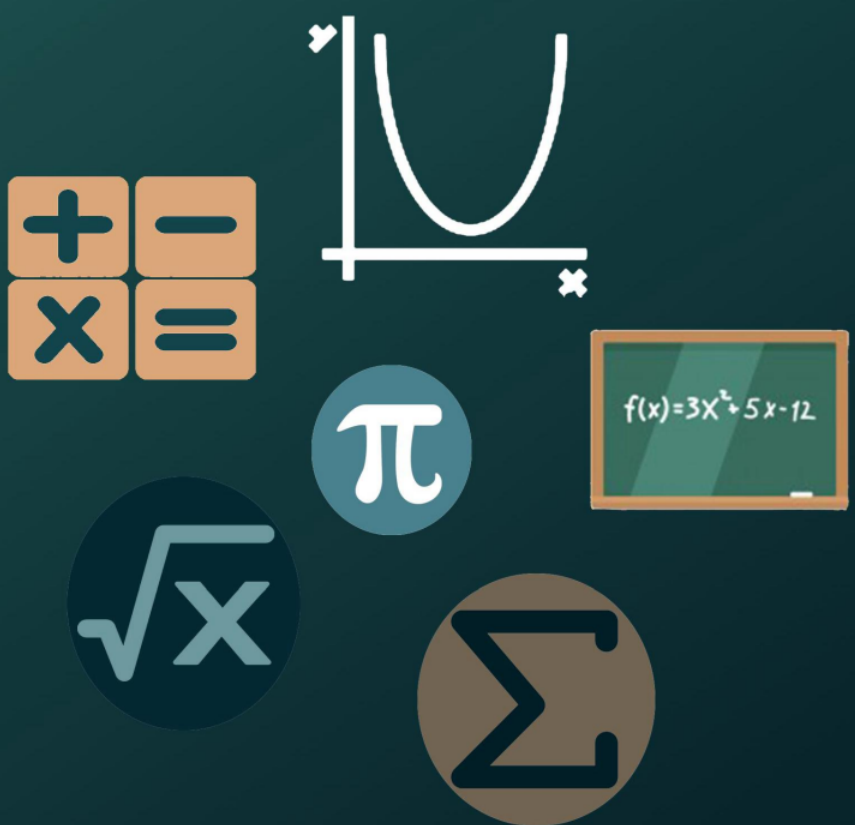
MATHEMATICS

Indefinite Integration



SHEET

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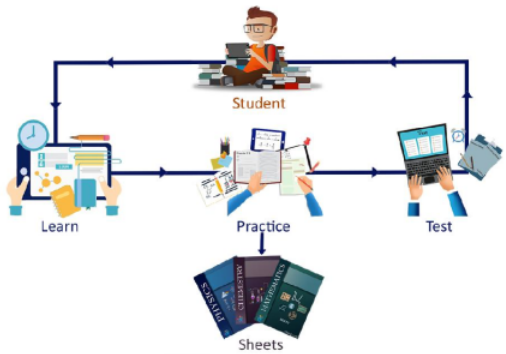
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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr. No.	Topics	Ex - 1	Ex - 1A	Ex - 2	Ex - 2A
1	Elementary Integrals	1 - 13	1 - 2, 6 - 8, 12 - 15		
2	Integration by Substitution - Part 1	14 - 28		1 - 2	
3	Integration by Substitution - Part 2	29 - 33			
4	Integration by Substitution - Part 3	34 - 39	3, 9	3 - 5	
5	Integration by Parts - Part 1	40 - 41	16	6 - 9	1 - 2
6	Integration by Parts - Part 2	42			18
7	Integration by Parts - Part 3	43 - 45		10	3 - 4
8	Integration of Trigonometric Functions - Part 1	46	4, 17	11 - 12	5 - 7
9	Integration of Trigonometric Functions - Part 2	47 - 48	10	13 - 15	19, 20
10	Integration by Partial Fraction	49 - 50	5	16	8
11	Integration of Algebraic Expressions		11, 18 - 19	17	15
12	Integrand Involving Even Powers of "x"	51 - 52		18 - 20	
13	Integration of Irrational Algebraic Expressions	53	20	21 - 22	21-24
14	Reduction Formula				14
15	Miscellaneous Integrals	54 - 55		23 - 25	16, 17
16	After Complete Chapter				

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Elementary Integrals

Q1. $\int \sqrt{1-\sin 2x}\,dx = \; x \in (0,\, \pi/4)$

(A) –sin x + cos x + c (B) sin x – cos x + c
(C) tan x + sec x + c (D) sin x + cos x + c

Q2. $\int \frac{dx}{\tan x + \cot x} =$

(A) $\frac{\cos 2x}{4} + c$ (B) $\frac{\sin 2x}{4} + c$

(C) $-\frac{\sin 2x}{4} + c$ (D) $-\frac{\cos 2x}{4} + c$

Q3. $\int (e^{a\ell nx} + e^{x\ell na})\,dx =$

(A) $x^{a+1} + \frac{a^x}{\ell n\, a} + c$

(B) $\frac{x^{a+1}}{a+1} + a^x \ell n\, a + c$

(C) $\frac{x^{a+1}}{a+1} + \frac{a^x}{\ell n\, a} + c$

(D) None of these

Q4. $\int \frac{\cos x - 1}{\cos x + 1} dx =$

(A) $2\tan\frac{x}{2} - x + c$ (B) $\frac{1}{2}\tan\frac{x}{2} - x + c$

(C) $x - \frac{1}{2}\tan\frac{x}{2} + c$ (D) $x - 2\tan\frac{x}{2} + c$

Q5. $\int \frac{dx}{1-\sin x} =$

(A) x + cos x + c (B) 1 + sin x + c
(C) sec x – tan x + c (D) sec x + tan x + c

Q6. $\int \sqrt{1+\sin\frac{x}{2}}\,dx =$

(A) $\frac{1}{4}\left(\cos\frac{x}{4} - \sin\frac{x}{4}\right) + c$

(B) $4\left(\cos\frac{x}{4} - \sin\frac{x}{4}\right) + c$

(C) $4\left(\sin\frac{x}{4} - \cos\frac{x}{4}\right) + c$

(D) $4\left(\sin\frac{x}{4} + \cos\frac{x}{4}\right) + c$

Q7. $\int \frac{\cos ec\,\theta - \cot\theta}{\cos ec\,\theta + \cot\theta} d\theta =$

(A) 2 cosec θ – 2 cot θ – θ + c
(B) 2 cosec θ – 2 cot θ + θ + c
(C) 2 cosec θ + 2 cot θ – θ + c
(D None of these

Q8. If $\int \frac{dx}{1+\sin x} = \tan\left(\frac{x}{2} + a\right) + b$, then a,b can be

(A) $a = \frac{\pi}{4},\ b = 3$

(B) $a = \frac{\pi}{4},\ b = 4$

(C) $a = \frac{\pi}{4},\ b =$ arbitrary constant

(D) $a = -\frac{\pi}{4},\ b =$ arbitrary constant

Q9. $\int \frac{\sin^3 x+\cos^3 x}{\sin^2 x\cos^2 x}dx =$

(A) tan x + cot x + c

(B) tan x – cot x + c

(C) cosec x – cot x + c

(D) sec x – cosec x + c

Q10. $\int \frac{dx}{\sqrt{1+x}+\sqrt{x}} =$

(A) $\frac{2}{3}(1+x)^{2/3}-\frac{2}{3}x^{2/3}+c$

(B) $\frac{3}{2}(1+x)^{2/3}+\frac{3}{2}x^{2/3}+c$

(C) $\frac{3}{2}(1+x)^{3/2}+\frac{3}{2}x^{3/2}+c$

(D) $\frac{2}{3}(1+x)^{3/2}-\frac{2}{3}x^{3/2}+c$

Q11. If $f'(x)=\frac{1}{x}+x$ and $f(1)=\frac{5}{2}$, then $f(x) =$ $(x>0)$

(A) $\ell n\,x+\frac{x^2}{2}+2$

(B) $\ell n\,x+\frac{x^2}{2}+1$

(C) $\ell n\,x-\frac{x^2}{2}+2$

(D) $\ell n\,x-\frac{x^2}{2}+1$

Q12. $\int(\sin^4 x-\cos^4 x)\,dx =$

(A) $-\frac{\cos 2x}{2}+c$ (B) $-\frac{\sin 2x}{2}+c$

(C) $\frac{\sin 2x}{2}+c$ (D) $\frac{\cos 2x}{2}+c$

Q13. $\int \frac{2x}{(2x+1)^2}dx =$

(A) $\frac{1}{2}\log|2x+1|+\frac{1}{2(2x+1)}+c$

(B) $\frac{1}{2}\log|2x+1|-\frac{1}{2(2x+1)}+c$

(C) $2\log|2x+1|+\frac{1}{2(2x+1)}+c$

(D) $2\log|2x+1|-\frac{1}{2(2x+1)}+c$

Integration by substitution-Part-1

Q14. $\int \frac{dx}{1+e^x} =$

(A) $\ell n\,(1+e^x)+c$ (B) $-\ell n\,(1+e^{-x})+c$

(C) $-\ell n\,(1-e^{-x})+c$ (D) $\ell n\,(e^{-x}+e^{-2x})+c$

Q15. $\int \frac{dx}{e^x+e^{-x}} =$

(A) $\tan^{-1}(e^{-x})+c$ (B) $\tan^{-1}(e^x)+c$

(C) $\ell n\,(e^x-e^{-x})+c$ (D) $\ell n\,(e^x+e^{-x})+c$

Q16. $\int \frac{e^{\sqrt{x}}\cos e^{\sqrt{x}}}{\sqrt{x}}dx =$

(A) $2\sin e^{\sqrt{x}}+c$ (B) $\sin e^{\sqrt{x}}+c$

(C) $2\cos e^{\sqrt{x}}+c$ (D) $-\sin e^{\sqrt{x}}+c$

Q17. $\int \frac{dx}{x+x\ell n\,x} =$

(A) $\ell n\,|1+\ell n\,x|+c$

(B) $\ell n\,\ell n\,|1+\ell n\,x|+c$

(C) $\ell n\,x+\ell n\,(\ell n\,x)+c$

(D) None of these

Q18. $\int \frac{x^{e-1}+e^{x-1}}{x^e+e^x}dx =$

(A) $\ell n\,(x^e+e^x)+c$ (B) $e\,\ell n\,(x^e+e^x)+c$

(C) $\frac{1}{e}\ell n(x^e+e^x)+c$ (D) None of these

Q19. $\int \frac{(1+\ell n\, x)^2}{x} dx =$

(A) $(1 + \ell n\ x)^3 + c$ (B) $3(1+ \ell n\ x)^3 + c$

(C) $\frac{1}{3}(1 + \ell n\ x)^3 + c$ (D) None of these

Q20. $\int \sec^p x \tan x\, dx =$ $(p \neq 0)$

(A) $\frac{\sec^{p+1} x}{p+1} + c$ (B) $\frac{\sec^p x}{p} + c$

(C) $\frac{\tan^{p+1} x}{p+1} + c$ (D) $\frac{\tan^p x}{p} + c$

Q21. $\int \frac{e^{\tan^{-1} x}}{1+x^2} dx =$

(A) $\ell n(1+x^2) + c$ (B) $\ell n\, e^{\tan^{-1} x} + c$

(C) $e^{\tan^{-1} x} + c$ (D) $\tan^{-1} e^{\tan^{-1} x} + c$

Q22. $\int \frac{\sqrt{\tan x}}{\sin x \cos x} dx =$

(A) $2\sqrt{\sec x} + c$ (B) $2\sqrt{\tan x} + c$

(C) $\frac{2}{\sqrt{\tan x}} + c$ (D) $\frac{2}{\sqrt{\sec x}} + c$

Q23. $\int e^{3\log x}(x^4+1)^{-1} dx$ is equal to

(A) $\frac{1}{4}\ell n\ (x^4+1) + C$ (B) $-\ell n(x^4+1) + C$

(C) $\ell n\ (x^4+1) + C$ (D) None of these

Q24. $\int \frac{\cos 2x + x + 1}{x^2 + \sin 2x + 2x} dx =$

(A) $\ell n\ |x^2 + \sin 2x + 2x| + c$

(B) $-\ell n\ |x^2 + \sin 2x + 2x| + c$

(C) $\frac{1}{2}\ell n\ |x^2 + \sin 2x + 2x| + c$

(D) None of these

Q25. $\int \frac{1}{x\cos^2(1+\ell n\, x)} dx =$

(A) $\tan(1 + \ell n\ x) + c$

(B) $\cot(1 + \ell n\ x) + c$

(C) $-\tan(1 + \ell n\ x) + c$

(D) $-\cot(1 + \ell n\ x) + c$

Q26. $\int \frac{x^2 \tan^{-1} x^3}{1+x^6} dx$ is equal to

(A) $\tan^{-1}(x^3) + c$

(B) $\frac{1}{6}(\tan^{-1} x^3)^2 + c$

(C) $-\frac{1}{2}(\tan^{-1} x^3)^2 + c$

(D) $\frac{1}{2}(\tan^{-1} x^2)^3 + c$

Q27. $\int x^x(1+\ell n\, x) dx$ is equal to

(A) $x^x + c$ (B) $x^{2x} + c$

(C) $x^x\ \ell n\ x + c$ (D) $\frac{1}{2}(1 + \ell n\ x)^2 + c$

Q28. $\int \frac{e^x(1+x)}{\cos^2(xe^x)}$ dx is equal to

(A) $\tan(xe^x) + c$ (B) $\tan^{-1}(xe^x) + c$

(C) $\sqrt{\tan(xe^x)} + c$ (D) None of these

Integration by substitution Part-2

Q29. $\int \sec x \ell n(\sec x + \tan x) dx =$

(A) $[\ell n\ (\sec x + \tan x)]^2 + c$

(B) $\frac{1}{2}[\ell n\ (\sec x + \tan x)]^2 + c$

(C) $\sec^2 x + \tan x \sec x + c$

(D) None of these

Q30. $\int \frac{\cot x}{\ell n \sin x} dx =$

(A) ℓn |ℓn sin x| + c

(B) ℓn |ℓn cosec x| + c

(C) 2ℓn |ℓn sin x| + c

(D) None of these

Q31. $\int x^2 \sec x^3 \, dx =$

(A) ℓn (sec x^3 + tan x^3)

(B) 3(sec x^3 + tan x^3)

(C) 1/3 ℓn (sec x^3 + tan x^3)

(D) None of these

Q32. $\int \frac{dx}{\sin(x-a)\sin(x-b)}$ is

(A) $\frac{1}{\sin(a-b)} \ell n \left|\frac{\sin(x-a)}{\sin(x-b)}\right| + c$

(B) $\frac{-1}{\sin(a-b)} \ell n \left|\frac{\sin(x-a)}{\sin(x-b)}\right| + c$

(C) $\ell n \sin(x-a)\sin(x-b) + c$

(D) $\ell n \left|\frac{\sin(x-a)}{\sin(x-b)}\right|$

Q33. The value of $\int \frac{dx}{x\sqrt{x^4-1}}$ is

(A) $\frac{1}{2}\sec^{-1} x^2 + k$

(B) $\ell n\, x\sqrt{x^4-1} + k$

(C) $x\, \ell n \sqrt{x^4-1} + k$

(D) $\ell n \sqrt{x^4-1} + k$

Integration by substitution Part-3

Q34. $\int \frac{1+x^2}{\sqrt{1-x^2}} dx =$

(A) $\frac{3}{2}\sin^{-1} x - \frac{1}{2} x\sqrt{1-x^2} + c$

(B) $\frac{3}{2}\sin^{-1} x + \frac{1}{2} x\sqrt{1-x^2} + c$

(C) $\frac{3}{2}\cos^{-1} x - \frac{1}{2} x\sqrt{1-x^2} + c$

(D) $\frac{3}{2}\cos^{-1} x + \frac{1}{2} x\sqrt{1-x^2} + c$

Q35. $\int \frac{\ell n(x+\sqrt{1+x^2})}{\sqrt{1+x^2}} dx =$

(A) $\frac{1}{2}[\ell n(x+\sqrt{1+x^2})]^2 + c$

(B) $\ell n(x+\sqrt{1+x^2}) + c$

(C) $\ell n(x+\sqrt{1+x^2}) + c$

(D) None of these

Q36. $\int \frac{dx}{(a^2+x^2)^{3/2}}$ is equal to

(A) $\frac{x}{(a^2+x^2)^{3/2}}$ + c (B) $\frac{x}{a^2(a^2+x^2)^{1/2}}$ + c

(C) $\frac{1}{a^2(a^2+x^2)^{1/2}}$ + c (D) None of these

Q37. $\int \sqrt{\frac{1+x}{1-x}} dx =$

(A) $-\sin^{-1} x - \sqrt{1-x^2} + c$

(B) $\sin^{-1} x + \sqrt{1-x^2} + c$

(C) $\sin^{-1} x - \sqrt{1-x^2} + c$

(D) $-\sin^{-1} x - \sqrt{x^2-1} + c$

Q38. $\int \frac{x+1}{\sqrt{1+x^2}}dx =$

(A) $\sqrt{1+x^2}+\tan^{-1}x+c$

(B) $\sqrt{1+x^2}-\ell n\left|x+\sqrt{1+x^2}\right|+c$

(C) $\sqrt{1+x^2}+\ell n\left|x+\sqrt{1+x^2}\right|+c$

(D) $\sqrt{1+x^2}+\ell n(\sec x+\tan x)+c$

Q39. $\int \frac{dx}{x^2+x+1}$ is equal to

(A) $\frac{\sqrt{3}}{2}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+C$

(B) $\frac{2}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+C$

(C) $\frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right)+C$

(D) None of these

Integration by Parts-Part-1

Q40. $\int \cos x.\ell n\tan\frac{x}{2}dx$ is equal to

(A) $\sin x.\ell n\tan\frac{x}{2}+x+C$

(B) $\sin x\ \ell n\tan\frac{x}{2}-x+C$

(C) $-\sin x\,\ell n\tan\frac{x}{2}-x+C$

(D) None of these

Q41. $\int (x-1)e^{-x}dx$ is equal to

(A) $-xe^x+C$ (B) xe^x+C

(C) $-xe^{-x}+C$ (D) $xe^{-x}+C$

Integration by Parts-Part-2

Q42. $\int \frac{x^2}{(x^2+a^2)(x^2+b^2)}dx$ is equal to

(A) $\frac{1}{(b^2-a^2)}\left(b\tan^{-1}\frac{x}{b}-a\tan^{-1}\frac{x}{a}\right)+C$

(B) $\frac{1}{b^2-a^2}\left(b\tan^{-1}\frac{x}{b}+a\tan^{-1}\frac{x}{a}\right)+C$

(C) $\frac{1}{b^2-a^2}\left(-b\tan^{-1}\frac{x}{b}+a\tan^{-1}\frac{x}{a}\right)+C$

(D) None of these

Integration by Parts-Part-3

Q43. $\int e^x\left(\frac{1-x}{1+x^2}\right)^2 dx$ is equal to

(A) $e^x\left(\frac{1-x}{1+x^2}\right)+C$ (B) $e^x\left(\frac{x-1}{1+x^2}\right)+C$

(C) $e^x.\frac{1}{1+x^2}+C$ (D) None of these

Q44. $\int \frac{e^x(1+\sin x)}{(1+\cos x)}dx$ is equal to

(A) $e^x\cot x+c$ (B) $\sin(\ell n\,x)+c$

(C) $e^x\tan x/2+c$ (D) $\ell n\tan x+c$

Q45. $\int \frac{e^{\tan^{-1}x}}{(1+x^2)}(1+x+x^2)dx$ is equal to

(A) $\frac{e^{\tan^{-1}x}}{1+x^2}+c$

(B) $e^{\tan^{-1}x}.(1+x^2)+c$

(C) $xe^{\tan^{-1}x}+c$

(D) None of these

Integration of Trigo Function-Part-1

Q46. $\int \sec^3 x\,dx$ is equal to

(A) $\frac{1}{2}\ell n\,|\sec x+\tan x|$

(B) $\frac{1}{2}\sec x\tan x+\frac{1}{2}\ell n\,|\sec x+\tan x|$

(C) $\frac{1}{2}\sec x\tan x$

(D) None of these

Q47. $\int \frac{\cot x}{\sqrt{\sin x}}dx$ is equal to

(A) $2\sqrt{\sin x}+C$

(B) $\frac{1}{2\sqrt{\sin x}}+C$

(C) $\frac{-2}{\sqrt{\sin x}}+C$

(D) $\frac{2}{\sqrt{\sin x}}+C$

Integration of Trigo Function-Part-2

Q48. $\int \frac{\sin x\cdot\cos x}{a\cos^2 x+b\sin^2 x}dx=$

(A) $\frac{1}{2(b-a)}\ell n\,|a\cos^2 x+b\sin^2 x|+c$

(B) $\frac{1}{(b-a)}\ell n\,|a\cos^2 x+b\sin^2 x|+c$

(C) $\frac{1}{2}\ell n\,|a\cos^2 x+b\sin^2 x|+c$

(D) None of these

Integration by Partial Fraction

Q49. $\int \frac{x^3-1}{x^3+x}dx$ is equal to

(A) $x+\ell n|x|+\frac{1}{2}\ell n\,(x^2+1)-\tan^{-1}x+C$

(B) $x-\ell n|x|+\frac{1}{2}\ell n(x^2+1)-\tan^{-1}x+C$

(C) $x-\ell n|x|-\frac{1}{2}\ell n(x^2+1)-\tan^{-1}x+C$

(D) None of these

Q50. $\int \frac{dx}{x(x^n+1)}$ is equal to

(A) $\ell n|x|-\frac{1}{n}\ell n|x^n+1|+C$

(B) $\frac{1}{n}\ell n\left|\frac{x^n+1}{x^n}\right|+C$

(C) $\ell n|x|+\frac{1}{n}\ell n\,|x^n+1|+C$

(D) None of these

Integrand Involving Even Powers of 'x'

Q51. $\int \frac{1}{1+x^4}dx$ is equal to

(A) $\frac{1}{2\sqrt{2}}\tan^{-1}\frac{x^2-1}{\sqrt{2}x}+\frac{1}{\sqrt{2}}\ell n\left|\frac{x^2-x\sqrt{2}+1}{x^2+x\sqrt{2}+1}\right|+C$

(B) $\frac{1}{2\sqrt{2}}\tan^{-1}\frac{x^2-1}{\sqrt{2}x}-\frac{1}{4\sqrt{2}}\ell n\left|\frac{x^2-x\sqrt{2}+1}{x^2+x\sqrt{2}+1}\right|+C$

(C) $\frac{1}{\sqrt{2}}\tan^{-1}\frac{x^2-1}{\sqrt{2}x}-\frac{1}{4\sqrt{2}}\ell n\left|\frac{x^2-x\sqrt{2}+1}{x^2+x\sqrt{2}+1}\right|+C$

(D) None of these

Q52. $\int \frac{x^2+1}{x(x^2-1)}dx$ is equal to

(A) $\ell n\left|\frac{x^2-1}{x}\right|+c$ (B) $-\ell n\left|\frac{x^2-1}{x}\right|+c$

(C) $\ell n\left|\frac{x}{x^2+1}\right|+c$ (D) $-\ell n\left|\frac{x}{x^2+1}\right|+c$

Integration of Irrational Algebric Expressions

Q53. $\int \frac{dx}{\sqrt{2x-x^2}} =$

(A) $\cos^{-1}(x-1)+c$ (B) $\sin^{-1}(x-1)+c$

(C) $\cos^{-1}(1+x)+c$ (D) $\sin^{-1}(1-x)+c$

Miscellaneous Integrals

Q54. Integral of $f(x)=\sqrt{1+x^2}$ with respect to x^2 is

(A) $\frac{2}{3}\frac{(1+x^2)^{3/2}}{x}+c$ (B) $\frac{2}{3}(1+x^2)^{3/2}+c$

(C) $\frac{2}{3}x(1+x^2)^{3/2}+c$ (D) None of these

Q55. $\int 5^{5^{5^x}}.5^{5^x}.5^x\,dx$ is equal to

(A) $5^{5^x}(\log 5)^{-3}+c$ (B) $5^{5^{5^x}}(\log 5)^3+c$

(C) $5^{5^{5^x}}(\log 5)^{-3}+c$ (D) None of these

Exercise - 1A *Concept Builder*

ONE OR MORE THAN ONE CORRECT TYPE

*Each question has FOUR options for correct answer(s). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If $\int(\sin 2x-\cos 2x)dx=\frac{1}{\sqrt{2}}\sin(2x-a)+b$, then which of the following can be CORRECT

(A) $a=\frac{\pi}{4}, b=0$

(B) $a=-\frac{3\pi}{4}, b=0$

(C) $a=\frac{5\pi}{4}, b=$ any constant

(D) $a=-\frac{5\pi}{4}, b=$ any constant

Q2. $\int x^{51}(\tan^{-1}x+\cot^{-1}x)\,dx =$

(A) $\frac{x^{52}}{52}(\tan^{-1}x+\cot^{-1}x)+c$

(B) $\frac{x^{52}}{52}(\tan^{-1}x-\cot^{-1}x)+c$

(C) $\frac{\pi x^{52}}{104}+\frac{\pi}{2}+c$

(D) $\frac{x^{52}}{52}+\frac{\pi}{2}+c$

Q3. $\int \frac{\sec^2 x\,dx}{\sqrt{\tan^2 x+4}} =$

(A) $\ell n\left|\tan x+\sqrt{\tan^2 x+4}\right|+c$

(B) $\frac{1}{2}\ell n\left|\tan x+\sqrt{\tan^2 x+4}\right|+c$

(C) $\ell n\left|\frac{1}{2}\tan x+\frac{1}{2}\sqrt{\tan^2 x+4}\right|+c$

(D) $\ell n\left|\tan x-\sqrt{\tan^2 x+4}\right|+c$

Q4. $\int \frac{1}{\sqrt{1+\sin x}}dx =$

(A) $\sqrt{2}\,\ell n\left|\tan\left(\frac{x}{4}+\frac{\pi}{8}\right)\right|$

(B) $\frac{1}{\sqrt{2}}\ell n\left|\tan\left(\frac{x}{4}+\frac{\pi}{8}\right)\right|$

(C) $-\sqrt{2}\,\ell n\left|\cos ec\left(\frac{x}{2}+\frac{\pi}{4}\right)+\cot\left(\frac{x}{2}+\frac{\pi}{4}\right)\right|$

(D) $\sqrt{2}\,\ell n\left|\cos ec\left(\frac{x}{2}+\frac{\pi}{4}\right)-\cot\left(\frac{x}{2}+\frac{\pi}{4}\right)\right|$

Q5. If $\int \frac{dx}{(x^2+1)(x^2+4)} = k\tan^{-1}x + \ell\tan^{-1}\frac{x}{2}+c$, then

(A) $k=\frac{1}{3}$

(B) $\ell=\frac{2}{3}$

(C) $k=-\frac{1}{3}$

(D) $\ell=-\frac{1}{6}$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions.Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph (Ques. 6 to 8)

y = f(x) is a polynomial function of leasst degree passing through (0, 1) and found to be increasing in the itnervals (1, 2) and (3, ∞) while it decreases in (–∞, 1) and (2, 3).

Q6. If f(1) = –8 then value of f(2) is

(A) – 3 (B) – 6 (C) – 20 (D) – 7

Q7. If f(1) = –8 then range of f(x) is

(A) [3, ∞) (B) [– 8, ∞)

(C) [– 7, ∞) (D) (– ∞, 6]

Q8. Equation f(x) = 0 has four solution then range in which leading coefficient of f(x) lies, is

(A) $\left(\frac{4}{9},\frac{1}{2}\right)$ (B) $\left[\frac{4}{9},\frac{1}{2}\right]$

(C) $\left(\frac{1}{3},\frac{1}{2}\right)$ (D) None

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q9. Column-I and column-II contains four entries each. Entry of column-I are to be matched with one or more than one entries of column-II.

Column-I	**Column-II**
(A) $\int\left(\frac{\cos x-\sin x}{\cos x+\sin x}\right)dx$	(P) $\ln\left\|\cos\left(x-\frac{\pi}{4}\right)\right\|+C$
(B) $\int\frac{\cos 2x}{1+\sin 2x}dx$	(Q) $(\cos^{-1}\sqrt{x})^2-\pi\cos^{-1}\sqrt{x}+C$

(C) $\int \frac{1-\sin 2x}{\cos 2x} dx$

(D) $\int \frac{\sin^{-1}\sqrt{x}}{\sqrt{x-x^2}} dx$

(R) $(\sin^{-1}\sqrt{x})^2 + C$

(S) $\ln|\cos x + \sin x| + C$

Q10. The antiderivative of

Column-I	**Column-II**
(A) $f(x) = \frac{1}{(a^2+b^2)-(a^2-b^2)\cos x}$ is	(P) $\frac{1}{ab}\tan^{-1}\left(\frac{a}{b}\tan\frac{x}{2}\right) + c$
(B) $f(x) = \frac{1}{a^2\sin^2 x + b^2\cos^2 x}$ is	(Q) $\frac{1}{a^2\sin\alpha}\tan^{-1}\left(\frac{\tan x}{\sin\alpha}\right) + c$, $\alpha = \cos^{-1}\frac{b}{a}$
(C) $f(x) = \frac{1}{a\cos x + b\sin x}$ is	(R) $\frac{1}{ab}\tan^{-1}\left(\frac{a}{b}\tan x\right) + c$
(D) $f(x) = \frac{1}{a^2 - b^2\cos^2 x}$ is ; $(a^2 > b^2)$	(S) $\frac{1}{\sqrt{a^2+b^2}}\log\left\lvert\tan\frac{1}{2}\left(x+\tan^{-1}\frac{a}{b}\right)\right\rvert + c$

Q11.

Column-I	**Column-II**
(A) $\int \frac{x^2(x^6+x^5-1)dx}{(2x^6+3x^5+2)^2}$	(P) $\frac{1}{2}(1 + x^{-2} + x^{-5})^{-2} + C$
(B) $\int \frac{(x^5+x^4+x^2)dx}{\sqrt{4x^7+5x^6+10x^4}}$	(Q) $\frac{-1}{6}(2x^3 + 3x^2 + 2x^{-3})^{-1} + C$
(C) $\int \frac{(2x^{12}+5x^9)dx}{(x^5+x^3+1)^3}$	(R) $x\left(\frac{x^3}{25}+\frac{x^2}{20}+\frac{1}{10}\right)^{\frac{1}{2}} + C$

(where C is the constant of integration.)

SUBJECTIVE TYPE

Q12. $\int \frac{(\sqrt{x}+1)(x^2-\sqrt{x})}{x\sqrt{x}+x+\sqrt{x}} dx$

Q13. $\int \left(\sqrt{x}+\frac{1}{\sqrt{x}}\right)^2 dx =$

Q14. $\int \cos^2 x/2\, dx =$

Q15. $\int \cos 3x \cos 5x\, dx$

Q16. Evaluate $\int x^2 e^x dx$

Q17. Evaluate $\int \frac{dx}{1+3\sin^2 x}$

Q18. $\int \frac{2x}{x^2+3x+2}dx$

Q19. $\int \frac{x+1}{x^2+x+1}dx$

Q20. $\int \frac{dx}{\sqrt{x^2+2x}}$

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. $\int \frac{dx}{\sqrt{x}\left(1+x^2\right)^{5/4}} =$

(A) $\frac{2x}{\left(1+x^2\right)^{1/4}}+C$ (B) $\frac{2\sqrt{x}}{\left(1+x^2\right)^{1/4}}+C$

(C) $\frac{2\sqrt{x}}{\sqrt{1+x^2}}+C$ (D) None of these

Q2. $\int \frac{\{f(x).\phi'(x)-f'(x).\phi(x)\}}{f(x).\phi(x)}\{\ell n\,\phi(x)-\ell n\,f(x)\}dx$ is equal to

(A) $\ell n\left|\frac{\phi(x)}{f(x)}\right|+c$

(B) $\frac{1}{2}\left\{\ell n\left|\frac{\phi(x)}{f(x)}\right|\right\}^2+c$

(C) $\frac{\phi(x)}{f(x)}\ell n\left|\frac{\phi(x)}{f(x)}\right|+c$

(D) None of these

Q3. $\int \sqrt{\frac{x}{a^3-x^3}}dx =$

(A) $\sin^{-1}\left(\frac{x}{a}\right)^{3/2}+c$

(B) $\frac{2}{3}\sin^{-1}\left(\frac{x}{a}\right)^{3/2}+c$

(C) $\frac{3}{2}\sin^{-1}\left(\frac{x}{a}\right)^{3/2}+c$

(D) $\frac{3}{2}\sin^{-1}\left(\frac{x}{a}\right)^{2/3}+c$

Q4. $\int \frac{1}{(x^2-1)\sqrt{x^2+1}}dx =$

(A) $\frac{1}{2\sqrt{2}}\ell n\left|\frac{\sqrt{1+x^2}+x\sqrt{2}}{\sqrt{1+x^2}-x\sqrt{2}}\right|+c$

(B) $\frac{1}{2\sqrt{2}}\ell n\left|\frac{\sqrt{1+x^2}-\sqrt{2}}{\sqrt{1+x^2}+\sqrt{2}}\right|+c$

(C) $\frac{1}{2\sqrt{2}}\ell n\left|\frac{\sqrt{1+x^2}-x\sqrt{2}}{\sqrt{1+x^2}+x\sqrt{2}}\right|+c$

(D) None of these

Q5. Let $f(x)=\int \frac{x^2dx}{(1+x^2)(1+\sqrt{1+x^2})}$ and f(0) = 0. Then f(1) is-

(A) $\ell n\,(1+\sqrt{2})$ (B) $\ell n\,(1+\sqrt{2})-\frac{\pi}{4}$

(C) $\ell n\,(1+\sqrt{2})+\frac{\pi}{4}$ (D) None of these

Q6. $\int e^{\sin x}.\frac{(x\cos^3 x-\sin x)}{\cos^2 x}dx$ is equal to

(A) $e^{\sin x}(\sec x-x)+C$

(B) $e^{\sin x}(x-\sec x)+C$

(C) $e^{\sin x}(\tan x-x)+C$

(D) None of these

Q7. If $\int\left(\ell n(\ell n x)+\frac{1}{(\ell n x)^2}\right)dx=x[f(x)-g(x)]+C,$ then

(A) $f(x)=\ell n\ (\ell n x), g(x)=\frac{1}{\ell n x}$

(B) $f(x)=\ell n x, g(x)=\frac{1}{\ell n x}$

(C) $f(x)=\frac{1}{\ell n x}, g(x)=\ell n\ (\ell n x)$

(D) None of these

Q8. If $\int x\ell n\left(1+\frac{1}{x}\right)dx=f(x).\ell n(x+1)$

$+g(x).x^2+Ax+c,$

then

(A) $f(x)=\frac{1}{2}x^2$ (B) g (x) = ln x

(C) A = 1 (D) None of these

Q9. If

$\int\frac{x^2}{(x\sin x+\cos x)^2}dx=\frac{f(x)}{x\sin x+\cos x}+\tan x+C,$

then

(A) $f(x)=\frac{x}{\cos x}$ (B) $f(x)=\frac{\cos x}{x}$

(C) $f(x)=\frac{-x}{\cos x}$ (D) None of these

Q10. The value of $\int e^x\frac{(x^3+x+1)}{(1+x^2)^{3/2}}dx$ is equal to

(A) $xe^x(1+x^2)^{5/2}+c$

(B) $xe^x(1+x^2)^{3/2}+c$

(C) $xe^x(1+x^2)^{-1/2}+c$

(D) None of thesez

Q11. If $\int\frac{1}{(\sin x+4)(\sin x-1)}dx$

$=A\frac{1}{\tan\frac{x}{2}-1}+B\tan^{-1}\{f(x)\}+c,$ then -

(A) $A=\frac{1}{5}, B=-\frac{2}{5\sqrt{15}}, f(x)=\frac{4\tan x+3}{\sqrt{5}}$

(B) $A=-\frac{1}{5}, B=\frac{1}{\sqrt{15}}, f(x)=\frac{4\tan\left(\frac{x}{2}\right)+1}{\sqrt{15}}$

(C) $A=\frac{2}{5}, B=-\frac{2}{5}, f(x)=\frac{4\tan x+1}{5}$

(D) $A=\frac{2}{5}, B=-\frac{2}{5\sqrt{15}}, f(x)=\frac{4\tan\frac{x}{2}+1}{\sqrt{15}}$

Q12. If $\int\frac{x+(\cos^{-1}3x)^2}{\sqrt{1-9x^2}}dx$

$=A\sqrt{1-9x^2}+B(\cos^{-1}3x)^3+c,$

where c is integration constant, then the values of A and B are

(A) $A=-\frac{1}{9}, B=-\frac{1}{9}$

(B) $A=-\frac{1}{9}, B=\frac{1}{9}$

(C) $A=\frac{1}{9}, B=\frac{1}{9}$

(D) None of the above

Q13. $\int \frac{\sin 2x}{\sin^4 x + \cos^4 x} dx =$

(A) $\cot^{-1}(\tan^2 x) + c$

(B) $\tan^{-1}(\tan^2 x) + c$

(C) $\cot^{-1}(\cot^2 x) + c$

(D) $\tan^{-1}(\cot^2 x) + c$

Q14. If $\int \frac{dx}{\sqrt{2ax - x^2}} = (f \circ g)(x) + C$, then

(A) $f(x) = \sin^{-1} x, g(x) = \frac{x+a}{a}$

(B) $f(x) = \sin^{-1} x, g(x) = \frac{x-a}{a}$

(C) $f(x) = \cos^{-1} x, g(x) = \frac{x-a}{a}$

(D) $f(x) = \tan^{-1} x, g(x) = \frac{x-a}{a}$

Q15. If $\int \frac{\sin^8 x - \cos^8 x}{1 - 2\sin^2 x \cos^2 x} dx = A \sin 2x + B$, then

(A) $A = -\frac{1}{2}$ (B) $A = \frac{1}{2}$

(C) $A = -1$ (D) $A = 1$

Q16. $\int \frac{(x+1)}{x\left(1 + xe^x\right)^2} dx = \ell n\left|-f(x)\right| + f(x) + C$,

then $f(x) =$

(A) $\frac{1}{x + e^x}$

(B) $\frac{1}{1 - xe^x}$

(C) $\frac{1}{\left(1 + xe^x\right)^2}$

(D) None of these

Q17. If $\int \frac{4e^x + 6e^{-x}}{9e^x - 4e^{-x}} dx = Ax + B\ell n\ (9e^{2x} - 4) + C$,

then A, B and C will be

(A) $A = -\frac{3}{2}, B = \frac{36}{35}, C = \text{constant}$

(B) $A = \frac{3}{2}, B = \frac{35}{36}, C = \frac{3}{2}\ell n\ 3 + c$

(C) $A = -\frac{3}{2}, B = -\frac{35}{36}, C = -\frac{3}{2}\ell n 3 + c$

(D) $A = -\frac{3}{2}, B = \frac{35}{36}, C = \text{constant}$

Q18. $\int \frac{(x^2 - 1)}{x\sqrt{x^4 + 3x^2 + 1}} dx$ is equal to

(A) $\ell n\left|x + \frac{1}{x} + \sqrt{x^2 + \frac{1}{x^2} + 3}\right| + C$

(B) $\ell n\left|x - \frac{1}{x} + \sqrt{x^2 + \frac{1}{x^2} - 3}\right| + C$

(C) $\ell n\left|x + \sqrt{x^2 + 3}\right| + C$

(D) None of these

Q19. $\int \frac{x^4 + 1}{x^6 + 1} dx$ is equal to

(A) $\tan^{-1} x + \frac{1}{3}\tan^{-1} x^3 + C$

(B) $\tan^{-1} x - \frac{1}{3}\tan^{-1} x^3 + C$

(C) $-\tan x - \frac{1}{3}\tan^{-1} x^3 + C$

(D) None of these

Q20. $\int \frac{dx}{\sin^6 x + \cos^6 x}$ is equal to

(A) $\tan^{-1}(\tan x + \cot x) + C$

(B) $\tan^{-1}(\cot x - \tan x) + C$

(C) $\tan^{-1}(\tan x - \cot x) + C$

(D) None of these

Q21. If $\int \frac{dx}{(1+x^2)\sqrt{1-x^2}} = -\frac{1}{\sqrt{2}}\sin^{-1}[f(x)] + C$, then

(A) $f(x) = \sqrt{\frac{1-x^2}{1+x^2}}$ (B) $f(x) = \sqrt{\frac{1+x^2}{1-x^2}}$

(C) $f(x) = \sqrt{\frac{x^2-1}{x^2+1}}$ (D) $f(x) = \frac{\sqrt{2}x}{\sqrt{1+x^2}}$

Q22. $\int \frac{x+2}{(x^2+3x+3)\sqrt{x+1}}\,dx =$

(A) $\frac{2}{\sqrt{3}}\tan^{-1}\frac{x}{\sqrt{3(x+1)}}$

(B) $\frac{1}{\sqrt{3}}\tan^{-1}\frac{x}{\sqrt{3(x+1)^2}}$

(C) $\frac{2}{\sqrt{3}}\tan^{-1}\frac{x}{\sqrt{x+1}}$

(D) None of these

Q23. If $\int g(x)\{f(x)+f'(x)\}dx = f(x)g(x)$, then $|g(x)|$ is :- (where, $f(x) \neq 0$)

(A) $e^{-x}e^c$

(B) $e^x e^c$

(C) $\ell n\ e^x e^c$

(D) $-\ell n\ e^{-x}e^c$

Q24. If $I = \int \frac{1}{2p}\sqrt{\frac{p-1}{p+1}}\,dp = f(p) + c$, then $f(p)$ is equal to

(A) $\frac{1}{2}\ell n\left|p - \sqrt{p^2-1}\right|$

(B) $\frac{1}{2}\cos^{-1} p + \frac{1}{2}\sec^{-1} p$

(C) $\ell n\sqrt{p+\sqrt{p^2-1}} - \frac{1}{2}\sec^{-1} p$

(D) None of these

Q25. $\int \{f(x)g''(x) - f''(x)g(x)\}dx$ is equal to

(A) $\frac{f(x)}{g'(x)}$

(B) $f'(x)g(x) - f(x)g'(x)$

(C) $f(x)g'(x) - f'(x)g(x)$

(D) $f(x)g'(x) + f'(x)g(x)$

Exercise - 2A — *Brain Booster*

ONE OR MORE THAN ONE CORRECT TYPE

Each question has FOUR options for correct answer(s). ***ONE OR MORE THAN ONE*** *of these four option(s) is (are) correct option(s). For each question, choose the correct option(s) to answer the question.*

Q1. If $\int \frac{bx\cos 4x - a\sin 4x}{x^2}dx = \frac{a\sin 4x}{x}$ then a and b may be

(A) $a = 1, b = 4$ (B) $a = -1, b = 4$

(C) $a = 1, b = \frac{1}{4}$ (D) $a = \frac{1}{4}, b = 1.$

Q2. If $\int(\cos^{-1}x+\cos^{-1}\sqrt{(1-x^2)})\,dx$

$= Ax + f(x)\sin^{-1}x$

$-2\sqrt{(1-x^2)}+c\ \forall\ x\in[-1,0)$, then-

(A) f(x) = x (B) f(x) = –2x

(C) $A=\frac{\pi}{4}$ (D) $A=\frac{\pi}{2}$

Q3. If the primitive of sin (ℓn x) is f(x) {sin g(x) – cos h(x)} + c (c being the constant of integration), then -

(A) $\lim_{x\to2} f(x)=1$ (B) $\lim_{x\to1}\frac{g(x)}{h(x)}=1$

(C) $g(e^3)=3$ (D) $h(e^5)=5$

Q4. $\int\frac{\sqrt{1-\sin x}}{1+\cos x}.e^{-x/2}\,dx$ is equal to

(A) $\sec\frac{x}{2}.e^{-x/2}+C$

(B) $-\sec\frac{x}{2}.e^{-x/2}+C$

(C) $\tan\frac{x}{2}.e^{-x/2}+C$

(D) None of these

Q5. If $\int\frac{\sin x}{\sin(x-\alpha)}dx = Ax + B\,\ell n\,\sin(x-\alpha)+C$

then

(A) $A=-\cos\alpha$ (B) $A=\cos\alpha$

(C) $B=-\sin\alpha$ (D) $B=\sin\alpha$

Q6. If $\int\frac{\cos 4x+1}{\cot x-\tan x}dx = k\cos 4x + c$ then

(A) k = –1/2

(B) k must be negative

(C) k = –1/4

(D) k = –1/8

Q7. If $I=\int\frac{\sin x+\sin^3 x}{\cos 2x}dx =$

$A\cos x + B\log|f(x)|+c$, then

(A) $A=\frac{1}{4}, B=\frac{1}{2}, f(x)=\frac{\sqrt{2}\cos x-1}{\sqrt{2}\cos x+1}$

(B) $A=\frac{1}{2}, B=\frac{3}{4\sqrt{2}}, f(x)=\frac{\sqrt{2}\cos x-1}{\sqrt{2}\cos x+1}$

(C) $A=\frac{1}{2}, B=\frac{3}{4\sqrt{2}}, f(x)=\frac{\sqrt{2}\cos x+1}{\sqrt{2}\cos x-1}$

(D) $A=\frac{1}{2}, B=-\frac{3}{4\sqrt{2}}, f(x)=\frac{\sqrt{2}\cos x-1}{\sqrt{2}\cos x+1}$

Q8. The value of the integral

$\int\frac{\cos^3 x+\cos^5 x}{\sin^2 x+\sin^4 x}dx$ is

(A) sinx – 6 $\tan^{-1}$ (sinx) + c

(B) $\sin x-2(\sin x)^{-1}+6\cot^{-1}(\sin x)+c$

(C) $\sin x-2(\sin x)^{-1}-6\tan^{-1}(\sin x)+c$

(D) $\sin x-2(\sin x)^{-1}+c$

PARAGRAPH TYPE

*This section contains **PARAGRAPHS**. Based on each paragraph, there are questions. Each question has FOUR options. **ONLY ONE** of these four options corresponds to the correct answer. For each question, choose the option corresponding to the correct answer.*

Paragraph # 1 (Ques. 9 to 11)

In calculating a number of integrals we had to use the method of integration by parts several times in succession. The result could be obtained more rapidly and in a more concise form by using the so-called generalized formula for integration by parts,

$$\int u(x)v(x)dx = u(x)v_1(x)-u'(x)v_2(x)+u''(x)v_3(x)$$

$$+....+(-1)^{n-1}u^{n-1}(x)v_n(x)-(-1)^{n-1}\int u^n(x)v_n(x)dx$$

where $v_1(x)=\int v(x)dx, v_2(x)=\int v_1(x)dx.....,v_n(x)=\int v_{n-1}(x)dx$

Of course, we assume that all derivatives and integrals appearing in this formula exist. The use of the generzlized formula for integration by parts is especially useful when calculating $\int P_n(x)Q(x)dx$, where $P_n(x)$ is polynomial of degree n and the factor Q(x)is such that it can be integrated successively n + 1 times.

Q9. If $\int(x^3-2x^2+3x-1)\cos 2x\,dx$

$=\frac{\sin 2x}{4}u(x)+\frac{\cos 2x}{8}v(x)+C$ then -

(A) $u(x)=x^3-4x^2+3x$

(B) $u(x)=2x^3-4x^2+3x$

(C) $v(x)=3x^2-4x+3$

(D) $v(x)=6x^2-8x+3$

Q10. If $\int(2x^3+3x^2-8x+1)\sqrt{2x+6}\,dx=\frac{\sqrt{2x+6}}{579}$ (2x + 6) f (x) + C then f(x) is equal to

(A) $x^3-6x^2-91x+297$

(B) $7x^3-3x^2-132x+597$

(C) $70x^3-45x^2-396x+897$

(D) None of these

Q11. If $\int(3x^2+x-2)\sin^2(3x+1)\,dx=-\frac{u(x)}{72}$

$\sin(6x+2)+\frac{v(x)}{72}\cos(6x+2)$

$+\frac{1}{2}x^3+\frac{1}{4}x^2-x+C.$

(A) $u(x)=3x^2+6x-13$

(B) $u(x)=18x^2+2x-13$

(C) $v(x)=3x+1$

(D) $v(x)=-(6x+1)$

Paragraph # 2 (Ques. 12 to 14)

Integrals of class of functions following a definite pattern can be found by the method of reduction and recursion. If the integral of member of the class having positive integral power can be found then the other members of the class can be found by successive application of the recursion formula. While deriving a recursion formula. We integrate by parts and try to bring back the original integral.

For example, $I_n=\int(\sin x)^n dx$ can be expressed as

$I_n=-\frac{(\sin x)^{n-1}\cos x}{n}+\frac{n-1}{n}I_{n-2}$, Which is the required reduction formula and is true for all $n\geq 2$.

Q12. If $I_n=\int\tan^n x\,dx$, then the value of I_3 can be equal to -

(A) $\sec^2 x + \log\cos x + c$

(B) $\sec^2 x - \log\cos x + c$

(C) $\frac{1}{2}\sec^2 x - \log\cos x + c$

(D) $\frac{1}{2}\sec^2 x + \log\cos x + c$

Q13. If $I_n=\int\sec^n x\,dx$, then the relation between I_5 and I_3 will be -

(A) $3I_5=-\sec^3 x\tan x-2I_3$

(B) $3I_5=\sec^3 x\tan x-3I_3$

(C) $4I_5=\sec^3 x\tan x+3I_3$

(D) $4I_5=\sec^3 x\tan x-3I_3$

Q14. If $I_n=\int x\sin^n x\,dx$, then the value of $I_3\Big|_{x=\frac{\pi}{2}}$ must be equal to -

(A) $\frac{2}{3}$ (B) $\frac{4}{9}$

(C) $\frac{7}{9}$ (D) 1

MATCH THE COLUMN TYPE

*Following questions contain statements given in two columns, which have to be matched. The statements in **Column-I** are labelled as A, B, C and D while the statements in **Column-II** are labelled as (P), (Q), (R) and (S). Any given statement in **Column-I** can have correct matching with **ONE OR MORE** statement(s) in **Column-II**.*

Q15. $\int f(x)dx$ when

Column-I	**Column-II**
(A) $f(x) = \frac{1}{(a^2+x^2)^{3/2}}$	(P) $c - \frac{1}{a}\sin^{-1}\frac{a}{\lvert x\rvert}$
(B) $f(x) = \frac{x^2}{\sqrt{a^2-x^2}}$	(Q) $\frac{a^2}{2}\sin^{-1}\frac{x}{a} - \frac{x}{2}\sqrt{a^2-x^2}+c$
(C) $f(x) = \frac{1}{(x^2-a^2)^{3/2}}$	(R) $c - \frac{x}{a^2\sqrt{x^2-a^2}}$
(D) $f(x) = \frac{1}{x\sqrt{x^2-a^2}}$	(S) $\frac{x}{a^2\sqrt{x^2+a^2}}+c$

Q16.

(A) If $f(x) = \underbrace{\ell n\,\ell n\,\ell n \ldots . \ell nx.}_{r-\text{times}}$ (P) 2

$\int \{xf_1(x)f_2(x).....f_{15}(x)^{-1}dx = f_k(x)+c$

where k is divisible by

(B) The value of (Q) 3

$$\int \frac{1-\cot^4 x}{\tan x+\cot^5 x}dx = \frac{1}{k}\ell n\,|\sin^k x+\cos^k x\,|+c$$

where k is greater than α, then α can be

(C) The value of integral (R) 5

$$\int \frac{(1-\cos x)^{8/7}}{(1+\cos x)^{15/7}}dx = \frac{1}{k}\left(\tan\frac{x}{2}\right)^K+c$$

where k is less than α, then α can be

(D) If $\int x^2e^{x/2}dx = e^{x/2}f(x)+c$ (S) 6

then minimum value of f(x) is

(T) 8

Q17. Let $I = \int \frac{e^x}{e^{4x}+1}dx$ and $J = \int \frac{e^{-x}}{e^{-4x}+1}dx$ Then for any arbitrary constant C, match the following

Column-I	**Column-II**
(A) I	(P) $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^x}\right)+C$
(B) J + I	(Q) $\frac{1}{2\sqrt{2}}\ln\left\lvert\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}\right\rvert+C$
(C) J – I	(R) $\frac{1}{2\sqrt{2}}\left(\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^x}\right)-\frac{1}{2}\ln\left\lvert\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}\right\rvert\right)+C$
(D) J	(S) $\frac{1}{2\sqrt{2}}\left(\tan^{-1}\left(\frac{e^{2x}-1}{\sqrt{2}e^x}\right)+\frac{1}{2}\ln\left\lvert\frac{e^{2x}-\sqrt{2}e^x+1}{e^{2x}+\sqrt{2}e^x+1}\right\rvert\right)+C$

SUBJECTIVE TYPE

Q18. Evaluate the indefinite integral

$$\int \cos 2x \ln\left(\frac{\cos x+\sin x}{\cos x-\sin x}\right)dx.$$

Q19. Evaluate the indefinite integral

$$\int \frac{1}{2+\sin 2x+\cos 2x}.$$

Q20. $\int \frac{1}{\sin x-3\cos x-1}dx$

Q21. $\int \ell n\left(\sqrt{1-x}+\sqrt{1+x}\right)dx$

Q22. $\int \sqrt{x+\sqrt{x^2+2}}.dx$

Q23. $\int \sqrt{\frac{1-\sqrt[3]{x}}{1+\sqrt[3]{x}}}\frac{dx}{x}$

Q24. $\int \frac{x^2-1}{x^2+1}.\frac{dx}{\sqrt{1+x^4}}$

Answer Key

Ex - 1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
D	D	C	D	D	C	A	D	D	D
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
A	B	A	B	B	A	A	C	C	B
21	**22**	**23**	**24**	**25**	**26**	**27**	**28**	**29**	**30**
C	B	A	C	A	B	A	A	B	A
31	**32**	**33**	**34**	**35**	**36**	**37**	**38**	**39**	**40**
C	A	A	A	A	B	C	C	B	B
41	**42**	**43**	**44**	**45**	**46**	**47**	**48**	**49**	**50**
C	A	C	C	C	B	C	A	B	A
51	**52**	**53**	**54**	**55**					
B	A	B	B	C					

Ex - 1A

1. B, D **2.** A, C **3.** A, C **4.** A, C, D **5.** A, D

6. D **7.** B **8.** A

9. (A) $\rightarrow$ (P,S); (B) $\rightarrow$ (P,S); (C) $\rightarrow$ (P,S); (D) $\rightarrow$ (Q,R)

10. (A) $\rightarrow$ (P); (B) $\rightarrow$ (R); (C) $\rightarrow$ (S); (D) $\rightarrow$ (Q)

11. (A) $\rightarrow$ (Q); (B) $\rightarrow$ (R); (C) $\rightarrow$ (P) **12.** $\frac{x^2}{2} - x + c$

13. $\frac{1}{2}x^2 + \log|x + 2x| + c$ **14.** $\frac{1}{2}(x + \sin x) + c$

15. $\frac{1}{2}\left[\frac{1}{8}\sin 8x + \frac{1}{2}\sin 2x\right] + c$ **16.** $x^2e^x - 2xe^x + 2e^x + c$

17. $\frac{1}{2}\tan^{-1}(2\tan x) + c$ **18.** $2\log\left|\frac{(x+2)^2}{x+1}\right| + c$

19. $\frac{1}{2}\log(x^2 + x + 1) + \frac{1}{\sqrt{3}}\tan^{-1}\left(\frac{2x+1}{\sqrt{3}}\right) + c$

20. $\log\left|x + 1 + \sqrt{x^2 + 2x}\right| + c$

Ex - 2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	B	B	C	B	B	A	D	C	C
11	**12**	**13**	**14**	**15**	**16**	**17**	**18**	**19**	**20**
D	A	B	B	A	D	D	A	A	C
21	**22**	**23**	**24**	**25**					
D	A	B	C	C					

EX - 2A

1. A, D **2.** B, D **3.** A, B, C, D **4.** B **5.** B, C

6. B, D **7.** C, D **8.** B, C **9.** B, D **10.** D

11. D **12.** D **13.** C **14.** C

15. (A) $\rightarrow$ (S); (B) $\rightarrow$ (Q); (C) $\rightarrow$ (R); (D) $\rightarrow$ (P)

16. (A) $\rightarrow$ (P, T); (B) $\rightarrow$ (P, Q, R); (C) $\rightarrow$ (R, S, T); (D) $\rightarrow$ (T)

17. (A) $\rightarrow$ (R); (B) $\rightarrow$ (P); (C) $\rightarrow$ (Q); (D) $\rightarrow$ (S)

18. $\frac{\sin 2x}{2}\ln\left|\frac{\cos x+\sin x}{\cos x-\sin x}\right|+\frac{1}{2}\ln|\cos 2x|+C$

19. $\frac{1}{\sqrt{2}}\tan^{-1}\left(\frac{1+\tan x}{\sqrt{2}}\right)+C$

20. $\frac{1}{3}\ln\left|\frac{\tan\frac{x}{2}-1}{\tan\frac{x}{2}+2}\right|+C$

21. $x\ell n\left(\sqrt{1-x}+\sqrt{1+x}\right)-\frac{1}{2}x+\frac{1}{2}\sin^{-1}x+c$

22. $\frac{1}{3}\left(x+\sqrt{x^2+2}\right)^{3/2}-2\left(x+\sqrt{x^2+2}\right)^{-\frac{1}{2}}+C$

23. $3\left[\ell n|u|-\ell n\left(1+\sqrt{1-u^2}\right)-\arcsin u\right]+C$, where $u=\sqrt[3]{x}$

24. $-\frac{1}{\sqrt{2}}\tan^{-1}\frac{\sqrt{2}x}{\sqrt{1+x^4}}$

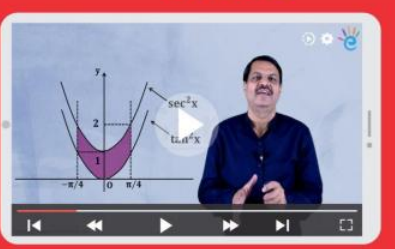
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